Fiscal Year:	FY 2011	**Task Last Updated:**	FY 07/13/2011
PI Name:	Bhattacharya, Sharmila		
Project Title:	Gravity and oxidative stress responses in Drosophila melanogaster		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Neurobiology		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Space Biology-FSB-Animal Physiology: NNH09ZTT003N
Start Date:	03/01/2010	**End Date:**	02/28/2013
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Marcu, Oana (SETI Institute)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The space environment of microgravity and radiation imposes stress conditions that affect the cellular and molecular responses of organisms and dictate their survival. A very fast response to stress is the production of reactive oxygen species (ROS), followed by intracellular oxidation. Space flight is associated with increased oxidative damage after return to 1g, suggesting that microgravity and/or the transition between gravity fields triggers a response. We propose to use a model organism used previously for space flight, the fruit fly Drosophila melanogaster. Studies in both hyper- and microgravity have already established many of the physiological alterations, and work from our own laboratory suggests that oxidative stress is implicated in the molecular response under both conditions. We will use hypergravity as a test environment for gravity alterations, and to dissect out both short-term molecular and cellular responses as well as long-term physiological outcomes. The goal is to provide a complete picture of the oxidative stress pathway in hypergravity, and to determine and select the key molecular functions and physiological read-outs to be tested in microgravity. The aims are 1. to quantify the short-term gravity-dependent cellular ROS production and alterations in oxidative enzyme gene expression; 2. to determine the key points of entry into the oxidative pathway by using mutants that show a physiological long-term altered response to hypergravity; 3. to determine the long-term effects of hypergravity over multiple generations. This approach will give a complete picture of the stress pathway, starting from fast cellular changes triggered by sudden changes in the gravity vector, to tissue-specific gene and enzyme activity alterations, all the way to physiological responses during animal lifetime and effects on population dynamics. Understanding the stress pathway in hypergravity opens the venue into looking at the opposite transition in the gravity load, from 1 g to microgravity during space flight, and will help predict the cellular and physiological response to other stresses imposed by the space environment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Some of the imaging tools developed to measure the behavioral changes for this project will be useful for a variety of behavioral studies in different organisms on Earth. The general issue of oxidative stress is also of great relevance to aging and stress on Earth, and the studies here therefore provide yet another paradigm in which to study and understand these molecular responses.
Task Progress:	1. Neuronal drivers and approach. The approach was to identify genes involved in the behavioral response to altered gravity (3g) by comparing fruit fly lines mutant for select genes with control lines. The RNA interference technique was used to obtain tissue-specific mutation in genes of interest. The initial experiments used the panneuronal driver elav-Gal4 to express doublestranded RNA in the entire brain. elav is present in both larvae and adults, and it gave a good starting point for an overall brain expression. Five other neuronal drivers were obtained and are being amplified and crossed. 2. Genotype screening for behavior alterations following hypergravity. Genes in five pathways (oxidative stress, longevity, behavior, autophagy and DNA repair, some overlapping) were identified as putative candidates for the response to altered gravity. Flies were obtained from the Vienna Drosophila RNAi Center (VDRC) and lines were amplified and crossed with the neuronal drivers.
Bibliography Type:	Description: (Last Updated: 03/02/2021)
Articles in Peer-reviewed Journals	Marcu O, Lera MP, Sanchez ME, Levic E, Higgins LA, Shmygelska A, Fahlen TF, Nichol H, Bhattacharya S. "Innate immune responses of Drosophila melanogaster are altered by spaceflight." PLoS One. 2011 Jan 11;6(1):e15361. http://dx.doi.org/10.1371/journal.pone.0015361 , Jan-2011
Articles in Peer-reviewed Journals	Inan O, Marcu O, Sanchez ME, Bhattacharya S, Kovacs GT. "A Portable System for Monitoring the Behavioral Activity of Drosophila." Journal of Neuroscience Methods, 2011. Submitted , Jan-2011